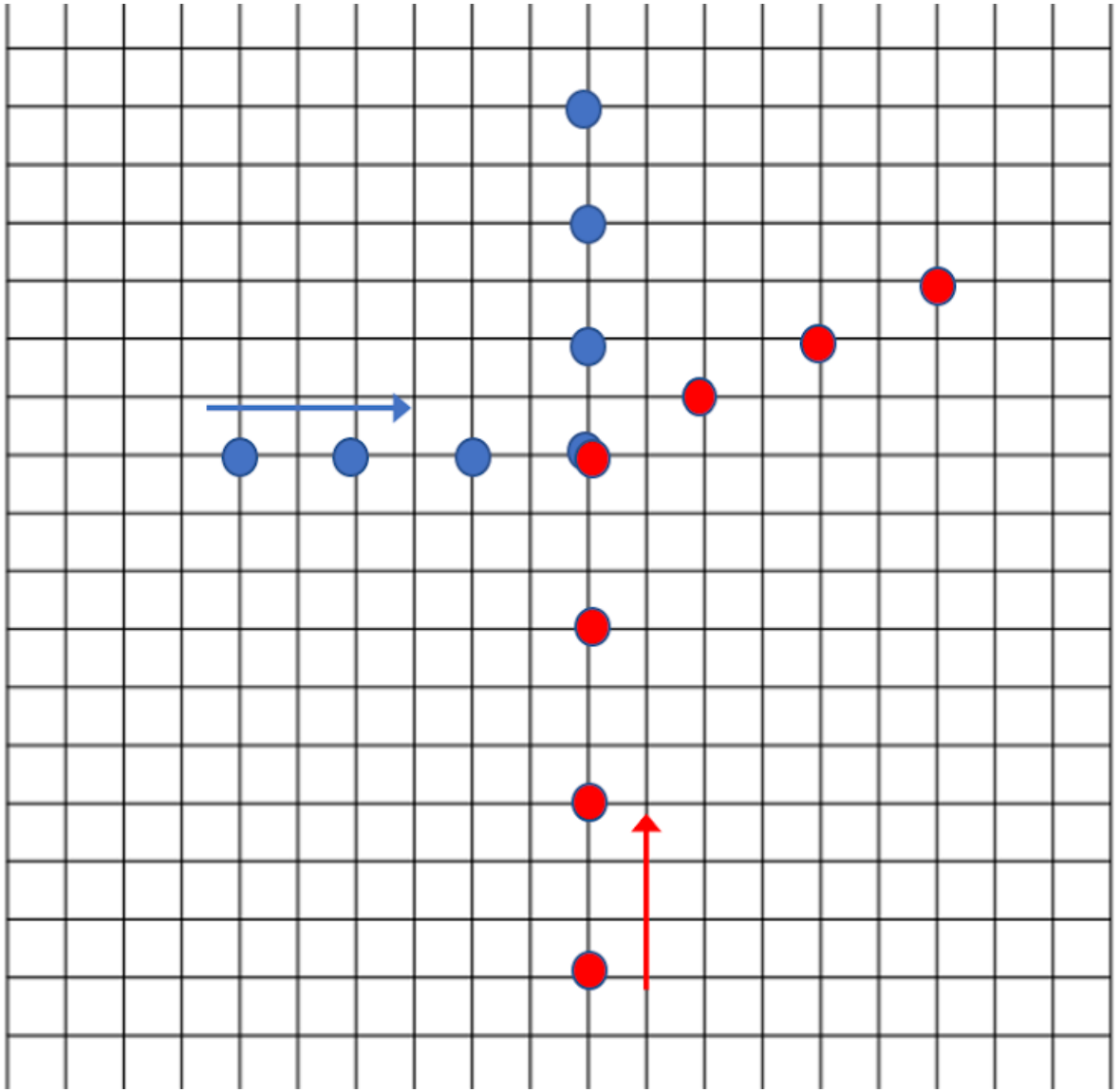
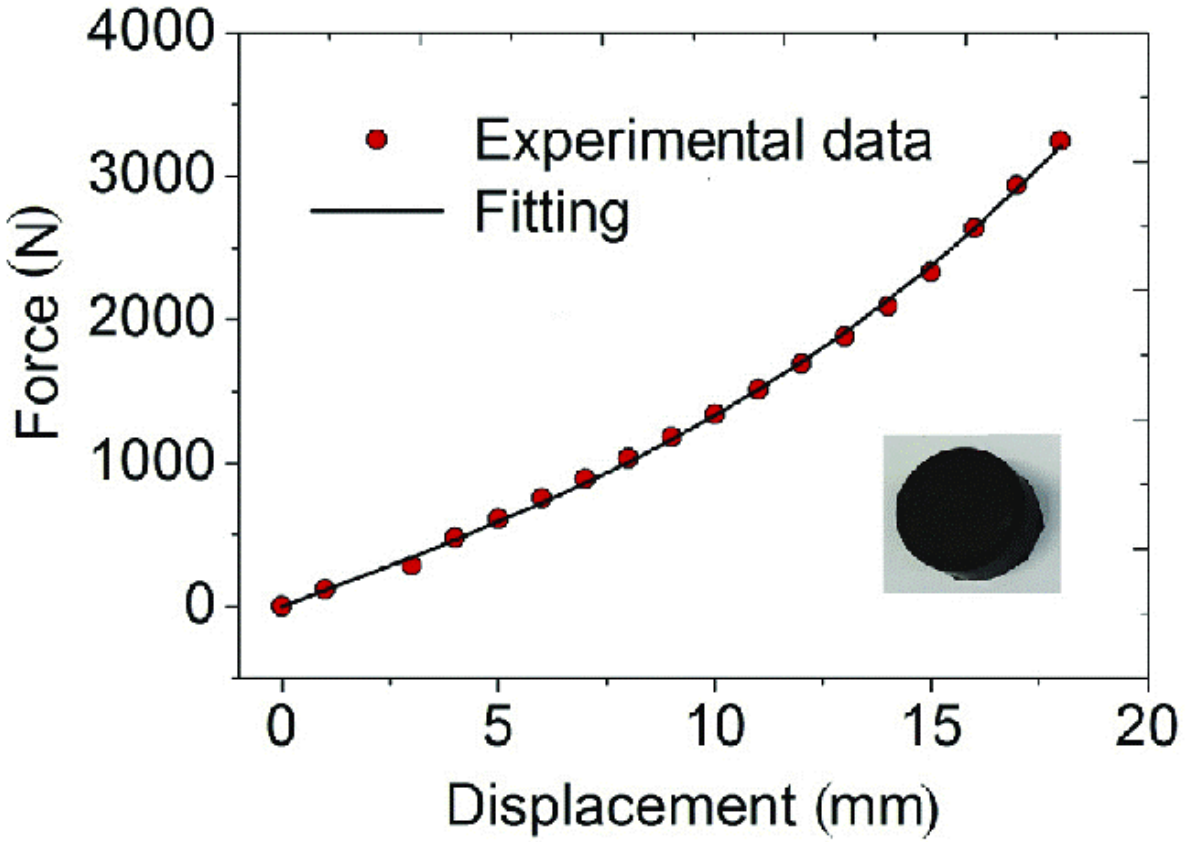
Two objects with equal mass leaves colored marks on a frictionless surface every time interval of T. They collide each other as shown in the figüre. What can be said about this collison?

10 puan



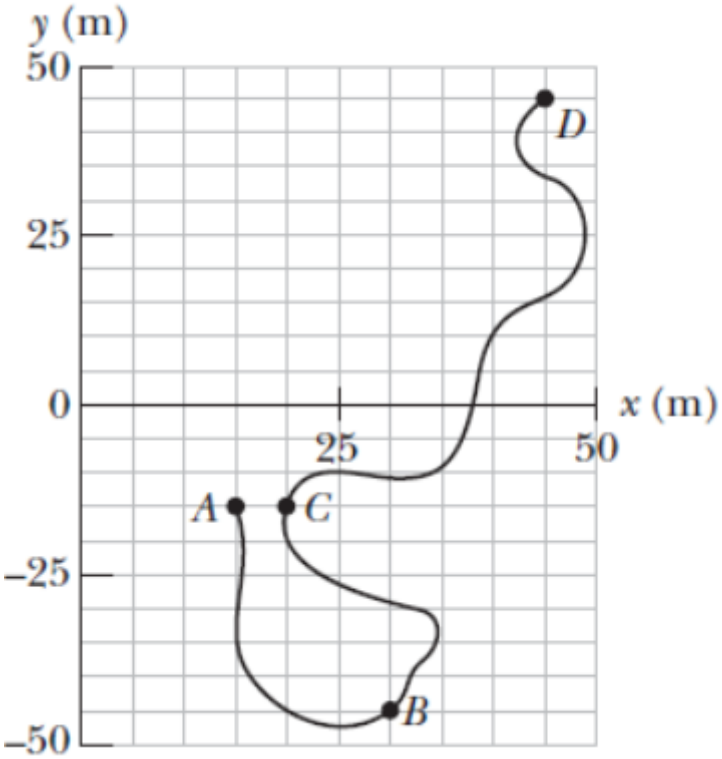
- The momentum of the red particle does not change
- The speed of the center of mass changes
- The momentum of the blue particle does not change
- This collision can not happen on a frictionless surface.
- The momentum is conserved in this collison

The figure shows Force-Displacement curve for natural rubber. What can be said about the rubber elasticity? 10 puan



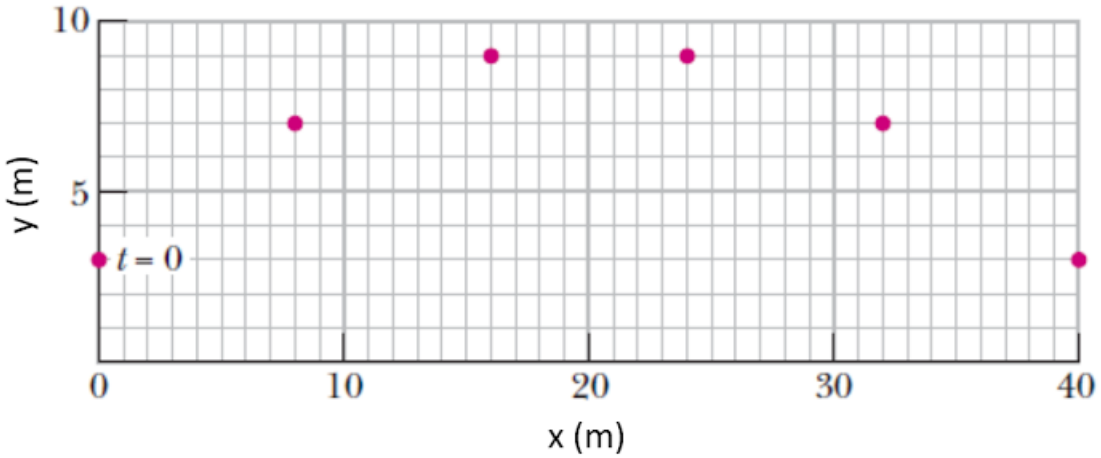
- All materials must behave the same way
- Natural rubber never obeys Hooke's law
- Natural rubber obeys Hooke's law for small strain values
- Natural rubber obeys Hooke's law
- This experiment must be wrong

The figure shows a path of a squirrel, moving from A (at time t=0), to point D (at 15.0 min). What is the magnitude of the velocity between point A and point D? 10 puan



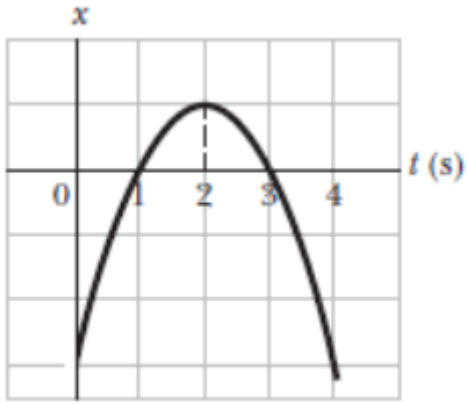
- 2.0 m/min
- 0.96 m/s
- 0.074 m/s
- 3.0 m/min
- 0.079 m/s

A stroboscopic picture of a ball is taken every 0.2 s. What is the speed of the ball when it reaches the highest y value on its path? 10 puan



- ◯ 20 m/s
- ◯ 40 m/s
- ◯ 80 m/s
- ◯ Can not be determined
- ◯ 50 m/s

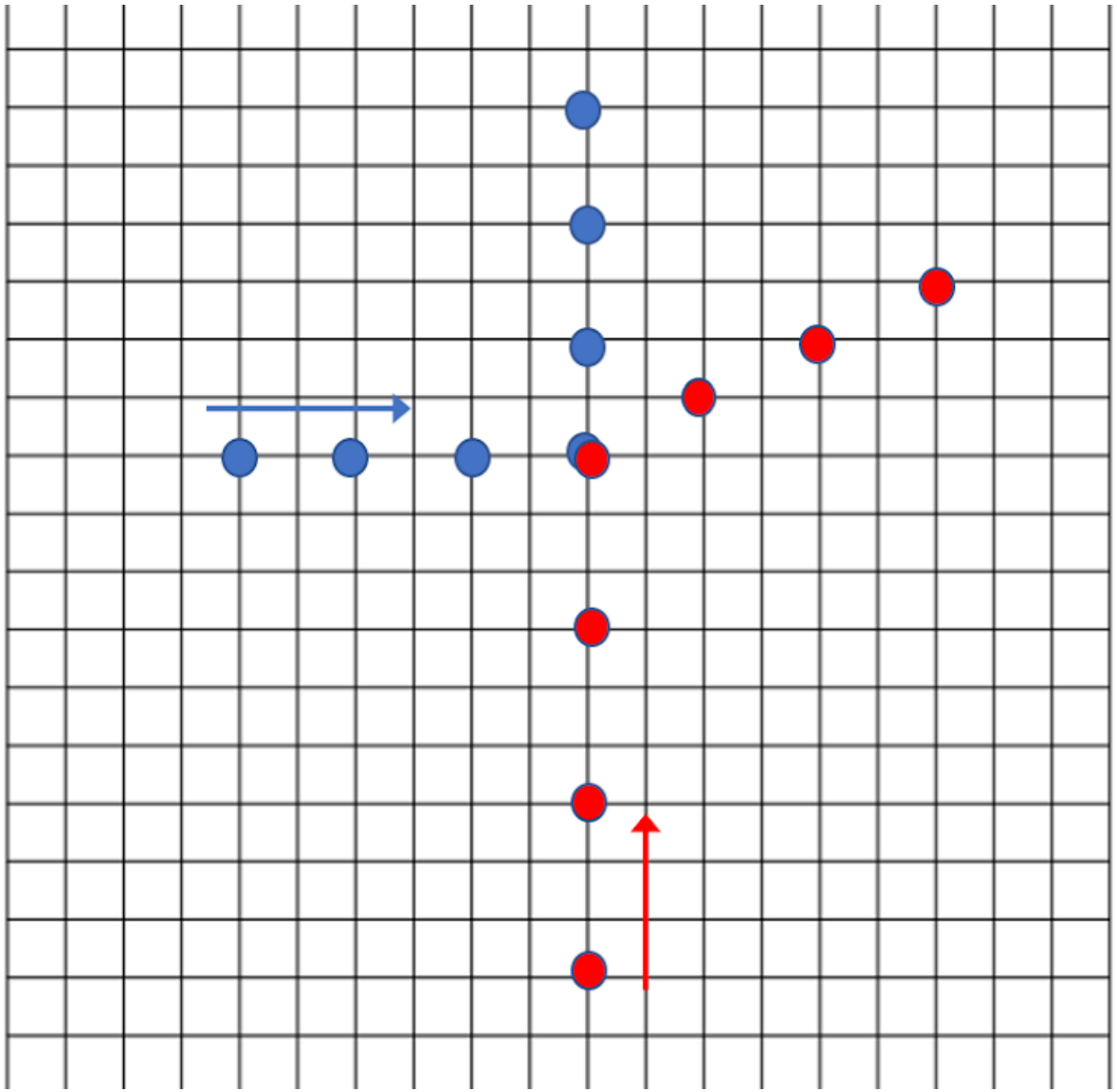
The graph is a particle’s position along x axis versus time. What are the signs of the particle's velocity at t=0, 1, 2, and 3 s. 10 puan



- -,0,+,+
- 0,+,0,+
- -,-,0,+
- -,0,+,0
- +,+,0,-

Two objects with equal mass leaves colored marks on a frictionless surface every time interval of T. They collide each other as shown in the figüre. What can be said about this collison?

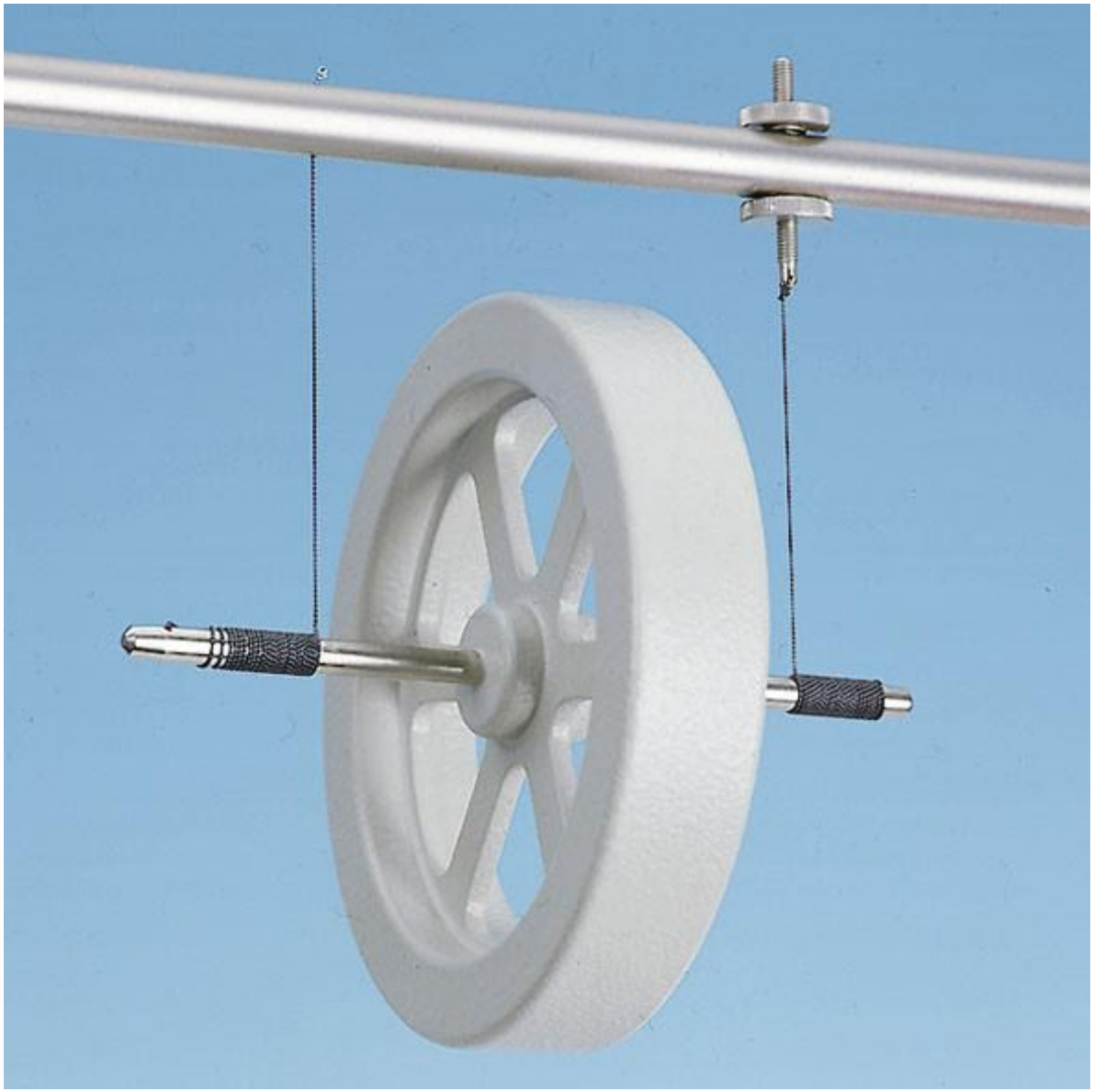
10 puan



- This is a completely inelastic scattering
- The total kinetic energy increases
- This is an elastic scattering
- The impulse on the blue object is the same as the impulse on the red object.
- This is an inelastic scattering

In a Maxwell's wheel experiment, the mass of the wheel is 450 gram, and the radius of the spindle going through the center of the wheel is 3 mm. A student lets the wheel fall 30 cm, and finds out that the speed of the center of mass is 0.15 m/s. What is the moment of inertia of this wheel? Take g=9.81 m/s^2.

10 puan



- 1.0 g.cm^2
- 1.0 kg.m^2
- 10 kg.m^2

- 2.0 kg.m^2
- 1.0x10^(-3) kg.m^2

Which one can be the unit of restring torque of a torsion axle? 10 puan

- kg.m^2
- kg.m^2/s
- kg.m/s^2
- none of them
- kg.m^2/s^2

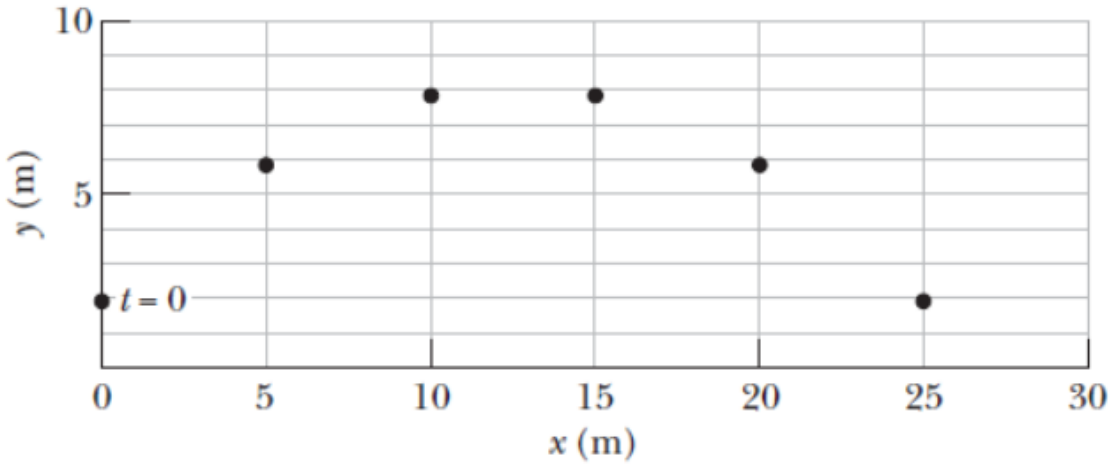
The period of a simple pendulum on a planet has been measured as a function of pendulum length. What can be said based on this data? 10 puan

Length (m)	T(s)
0.80	1.77
1.00	2.10
1.25	2.24
1.50	2.44
1.75	2.66
2.01	2.84

- ◯ This pendulum may be on Earth
- ◯ This data can not belong to a simple pendulum.
- ◯ The frequency of the pendulum increases with its length
- ◯ The free-fall acceleration on this plane is about 20.0 m/s^2.

- ◯ The free-fall acceleration is about 5.00 m/s^2 on this planet

On a distant planet, a particle is fired with an initial velocity, and its x and y positions are recorded every 0.5 s, as shown in the figure. What is the magnitude of the free-fall acceleration on this planet? 10 puan



- ◯ 10 m/s^2
- ◯ 4.0 m/s^2
- ◯ 8.0 m/s^2
- ◯ Not enough information available
- ◯ 6.0 m/s^2

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